EXPERIMENT 4

Single Sign-On (SSO) using Auth0

Step-by-Step Practice Guide

Aim: Implement Single Sign-On (SSO) for a sample application using Auth0 so a user authenticates once and can access multiple applications/resources without repeated login. Validate authentication and user profile retrieval.

Recommended practice setup

```
App A  ->  http://localhost:5173
App B  ->  http://localhost:5174
Identity Provider  ->  Auth0
```

Important: The lab record mixes ports 3000/3001 (common with Create React App) and 5173 (Vite). This guide consistently uses Vite with ports 5173 and 5174 so the procedure is easier to practice.

Before You Start

- Auth0 account (free tier)
- Node.js 16 or above and npm
- VS Code or another code editor
- A browser
- Internet connection for Auth0 login

Security note: Use your own Auth0 Domain and Client ID. Do not copy another student's credentials or secrets from a lab record.

Step 1: Check Node.js

Open Command Prompt or the VS Code terminal and run:

```
node -v
npm -v
```

Make sure Node.js is version 16 or above.

Step 2: Create or Sign In to an Auth0 Account

1. Open Auth0 in your browser and sign in or create an account.
2. Open the Auth0 Dashboard.
3. Create/select your tenant if Auth0 asks for one.

Step 3: Create Application A in Auth0

4. Go to Applications -> Applications.
5. Click Create Application.
6. Name it App A.
7. Choose Single Page Web Applications.
8. Click Create.

Step 4: Create the React Project for App A

In the terminal:

```
npm create vite@latest app-a

# Choose:
React
JavaScript

cd app-a
```

```
npm install
npm install @auth0/auth0-react
```

Step 5: Run App A

```
npm run dev
```

Vite normally starts the app at:

```
http://localhost:5173
```

Keep it open: Leave this terminal running while you configure Auth0.

Step 6: Configure App A URLs in Auth0

Go to Auth0 -> Applications -> App A -> Settings. Under Application URIs, set:

```
Allowed Callback URLs:   http://localhost:5173
Allowed Logout URLs:     http://localhost:5173
Allowed Web Origins:     http://localhost:5173
```

If your lab specifically uses /callback: Use http://localhost:5173/callback as the callback URL only if your code is written to handle that route.

Click Save Changes.

Step 7: Copy the Auth0 Domain and Client ID

In App A settings, copy these two values:

```
Domain
Client ID
```

Do not use Client Secret in a browser SPA: The React application only needs the Auth0 Domain and Client ID for this login flow.

Step 8: Create .env.local

Inside the app-a folder, create a file named .env.local and add:

```
VITE_AUTH0_DOMAIN=YOUR_AUTH0_DOMAIN
VITE_AUTH0_CLIENT_ID=YOUR_CLIENT_ID
```

After creating or changing .env.local, restart the Vite server:

```
Ctrl + C
npm run dev
```

Step 9: Add Auth0Provider in src/main.jsx

Replace src/main.jsx with:

```
import React from 'react'
import ReactDOM from 'react-dom/client'
import { Auth0Provider } from '@auth0/auth0-react'
import App from './App.jsx'

ReactDOM.createRoot(document.getElementById('root')).render(
  <Auth0Provider
    domain={import.meta.env.VITE_AUTH0_DOMAIN}
    clientId={import.meta.env.VITE_AUTH0_CLIENT_ID}
    authorizationParams={{
      redirect_uri: window.location.origin
    }}
  >
    <App />
  </Auth0Provider>
)
```

Step 10: Create the Login and Logout Interface

Replace src/App.jsx with this simple practice code:

```
import { useAuth0 } from "@auth0/auth0-react";

function App() {
  const {
    loginWithRedirect,
    logout,
    user,
    isAuthenticated,
    isLoading
  } = useAuth0();

  if (isLoading) {
    return <h2>Loading...</h2>;
  }

  return (
    <div>
      <h1>Application A</h1>

      {!isAuthenticated && (
```

```
      <button onClick={() => loginWithRedirect()}>
        Login
      </button>
    )}

    {isAuthenticated && (
      <>
        <h2>Welcome {user.name}</h2>
        <p>{user.email}</p>
        <button
          onClick={() =>
            logout({
              logoutParams: {
                returnTo: window.location.origin
              }
            })
          }
        >
          Logout
        </button>
      </>
    )}
  </div>
 );
}

export default App;
```

Step 11: Test App A Login

9. Open http://localhost:5173.
10. Click Login.
11. Auth0 redirects you to its login page.
12. Sign up/sign in or use a configured social login such as Google.
13. If an authorization/consent page appears, accept it.
14. Auth0 redirects you back to App A.
15. Confirm your name/email is displayed.

At this point: You have demonstrated authentication and user profile retrieval. The next steps demonstrate actual SSO across two applications.

Step 12: Create Application B in Auth0

16. Go to Applications -> Create Application.
17. Name it App B.
18. Choose Single Page Web Applications.
19. Create the app.

Configure these URLs for App B:

```
Allowed Callback URLs:   http://localhost:5174
Allowed Logout URLs:     http://localhost:5174
Allowed Web Origins:     http://localhost:5174
```

Click Save Changes.

Step 13: Create the React Project for App B

```
npm create vite@latest app-b

# Choose:
React
JavaScript

cd app-b
npm install
npm install @auth0/auth0-react
```

Step 14: Run App B on Port 5174

```
npm run dev -- --port 5174
```

Open:

```
http://localhost:5174
```

Step 15: Configure App B Environment Variables

Create app-b/.env.local and use App B's Client ID:

```
VITE_AUTH0_DOMAIN=YOUR_AUTH0_DOMAIN
VITE_AUTH0_CLIENT_ID=APP_B_CLIENT_ID
```

Expected: The Auth0 Domain is normally the same for both apps because they belong to the same tenant. The Client ID is different for each application.

Step 16: Copy the Auth0 Code into App B

Use the same main.jsx and App.jsx logic from App A. Change the heading in App.jsx to:

```
<h1>Application B</h1>
```

Step 17: Test Single Sign-On

20. Open http://localhost:5173 and log in to App A.
21. Do NOT log out.
22. Open a new browser tab.
23. Open http://localhost:5174.
24. Click Login in App B.
25. Auth0 should recognize the existing Auth0 session and authenticate you without requiring the full username/password login again.

SSO FLOW

```
        AUTH0
          |
      User Login
          |
    +-----+-----+
    |           |
    v           v
 App A       App B
 5173        5174
```

Step 18: What to Show the Examiner

26. Auth0 Dashboard
27. App A settings and allowed URLs
28. Auth0 Domain + Client ID (do not expose secrets)
29. App A before login
30. Auth0 login screen
31. Successful redirect back to App A
32. User name/email displayed
33. App B on port 5174
34. App B login succeeding without re-entering full credentials

Step 19: Short Explanation for Viva

Say this: The user authenticates with Auth0 once. Auth0 maintains the authentication session. When the same user accesses another application configured with the same Auth0 identity provider, Auth0 can authenticate the user without requiring the credentials again. This demonstrates Single Sign-On.

Step 20: Expected Result

```
Authentication          [OK]
User profile retrieval  [OK]
Auth0 login             [OK]
Logout                  [OK]
Single Sign-On          [OK]
Two applications        [OK]
```

Result

The application was successfully implemented with Single Sign-On (SSO) using Auth0 as the identity provider. A single authentication allowed access to multiple applications without repeated login, and the authenticated user profile was successfully retrieved in the React application.

Quick Troubleshooting

Callback URL mismatch: Make sure the URL in Auth0 exactly matches the URL used by your Vite app, including the port.

Environment variables are undefined: Confirm the variables start with VITE_, then restart npm run dev.

App B asks for credentials again: Make sure both applications use the same Auth0 tenant/domain and that you did not log out of Auth0 between tests.

Port already in use: Stop the old Vite process or run App B using npm run dev -- --port 5174.

Login returns an Auth0 error: Recheck Allowed Callback URLs, Allowed Logout URLs, and Allowed Web Origins in the Auth0 application settings.

Prepared from the uploaded Experiment 4 Auth0 lab record and reorganized into a practice-friendly sequence.